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Liu, Palcoux, and Wu ask in Question 6.15 of [1] whether every minimizer of the Donoho–Stark uncertainty principle in a fusion bialgebra is a bishift of a biprojection. Their Theorem 6.16 proves this when the dual has Young’s property. We give an affirmative answer for every canonical fusion bialgebra, in arbitrary dimension, without dual Young’s property. The result is invariant under every product-one gauge transformation, hence covers exactly the normalized setting in which the displayed uncertainty bound has sharp right-hand side one. This is a candidate full solution, likely valid pending expert review.

Question 6.15 occurs on PDF page 22 of [1]. The following is a real crop of the original arXiv PDF and includes the conditional theorem immediately following it.

Hence w is a right shift of $\mathcal{R}(\mathfrak{F}^{-1}(P))$. □

Corollary 6.14. *Let $(\mathcal{A}, \mathcal{B}, \mathfrak{F}, d, \tau)$ be a fusion bialgebra. Then a left shift of a biprojection is a right shift of a biprojection.*

Proof. It follows from Lemma 6.7 and Proposition 6.13. $\square$

Question 6.15. *Are the minimizers of the Donoho-Stark uncertainty principle bishifts of biprojections?*

Theorem 6.16. *Let $(\mathcal{A}, \mathcal{B}, \mathfrak{F}, d, \tau)$ be a fusion bialgebra. Suppose that the dual has Young's property. Then the minimizers of the Donoho-Stark uncertainty principle are bishifts of biprojections.*

Proof. The proof is similar to the proof of Proposition 6.13. We leave the details to the reader. $\square$

Theorem 6.17 (Exact inverse sum set theorem). *Let $(\mathcal{A}, \mathcal{B}, \mathfrak{F}, d, \tau)$ be a fusion bialgebra and P, Q projections in $\mathcal{A}$. Then the following are equivalent:*

- (1) $\mathcal{S}(P * Q) = \mathcal{S}(P)$;
- (2) $\frac{1}{\mathcal{S}(Q)} P * Q$ is a projection;
- (3) there is a biprojection B such that $Q \leq B_h$ and $P = \mathcal{R}(x * B)$ for some $x \neq 0$ in $\mathcal{A}$, where B_h is a right shift of B

Proof. (1) $\Rightarrow$ (2) By Theorem 5.19, we have that $\mathcal{S}(P * Q) \geq \mathcal{S}(P)$. By the assumption and the proof of Theorem 5.19 we have

The question reads:

“Are the minimizers of the Donoho–Stark uncertainty principle bishifts of biprojections?”

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CANONICAL NOTATION AND NORMALIZATION

Let $(\mathcal{A}, \mathcal{B}, \mathcal{F}, d, \tau)$ be a canonical fusion bialgebra. We use the canonical $\mathbb{R}_+$ -basis

$$\{x_1 = 1, x_2, \dots, x_m\} \subset \mathcal{B}, \quad x_j x_k = \sum_r N_{j,k}^r x_r,$$

where $N_{j,k}^r \geq 0$, $x_j^* = x_{j^*}$, and $d_j := \|x_j\| = d_{j^*}$. The dimension character satisfies

$$(1) \quad d_j d_k = \sum_r N_{j,k}^r d_r.$$

The basis is orthonormal for $L^2(\mathcal{B}, \tau)$ and obeys Frobenius reciprocity

$$(2) \quad N_{j,k}^r = N_{j^*,k}^r.$$

We use the same symbols x_j for their inverse Fourier transforms in $\mathcal{A}$. Then

$$p_j = d_j x_j$$

are the minimal projections of the abelian algebra $\mathcal{A}$, and $d(p_j) = d_j^2$.

For $0 \neq w \in \mathcal{A}$, put

$$\mathcal{S}(w) = d(\mathcal{R}(w)), \quad \mathcal{S}(\mathcal{F}w) = \tau(\mathcal{R}(\mathcal{F}w)).$$

The source proves $\mathcal{S}(w)\mathcal{S}(\mathcal{F}w) \geq 1$ and proves that equality is equivalent to w being an extremal bi-partial isometry [1, Section 6].

The canonical restriction loses no normalized cases. Indeed, the source proves that every fusion bialgebra is gauge-equivalent to a canonical one [1, Proposition 2.15 and Theorem 2.16]. Under the gauge

$$(\mathcal{F}, d, \tau) \mapsto (\lambda_1^{1/2} \lambda_2^{-1/2} \mathcal{F}, \lambda_1 d, \lambda_2 \tau),$$

the support product is multiplied by $\lambda_1 \lambda_2$. Thus its sharp lower constant remains one precisely for product-one gauges. Such a gauge has the form

$$(3) \quad (\mathcal{F}, d, \tau) \mapsto (\lambda \mathcal{F}, \lambda d, \lambda^{-1} \tau).$$

It leaves the equality cases and all support projections unchanged; it also rescales each side of the right-shift equation by λ and each side of the dual-shift equation by λ^{-1} . Hence both “minimizer” and “bishift” are invariant under (3).

PROOF INTUITION

Normalize a minimizer so that it has unit modulus on its support in the minimal projections p_j . Its Fourier transform is a sum of fusion-basis operators whose triangle-inequality bound is exactly attained. Equality is rigid: on the initial space of the Fourier partial isometry, every summand x_j must attain its norm and all summands must point in the same operator direction.

The pairwise products $x_j x_k^*$ therefore act as scalars of maximal modulus on the Fourier range projection. Positivity of the fusion coefficients forces

every constituent of every such product to do the same. These constituents generate a fusion subalgebra H . A coefficient calculation with the Fourier range projection shows that its phased regular element has exactly the same dimension as the original support. A second use of the same rigidity shows that the support is closed under left multiplication by H , so it is a fusion-theoretic coset. The ordinary and dual convolution formulas then give, respectively, the required right shift and Fourier-side shift. No inequality on the dual is used.

A SATURATED-TRIANGLE LEMMA

Lemma 1 (Rigidity of a positive fusion expansion). *Let $z = \sum_r c_r x_r$ with $c_r \geq 0$, and put $M = \sum_r c_r d_r$. Let P be a nonzero projection and let U be a partial isometry with $U^*U = P$. If, for a phase ω ,*

$$zP = M\omega U,$$

then, whenever $c_r > 0$,

$$x_r P = d_r \omega U.$$

Proof. For a unit vector $\xi \in P$, the operator-norm bounds $\|x_r\| = d_r$ give

$$M = \|z\xi\| \leq \sum_r c_r \|x_r \xi\| \leq \sum_r c_r d_r = M.$$

Equality holds throughout. Equality in the last inequality makes every x_r with $c_r > 0$ attain norm d_r on ξ , and equality in the vector triangle inequality makes all $c_r x_r \xi$ point in the direction $\omega U\xi$. Their prescribed lengths then give $x_r \xi = d_r \omega U\xi$. This holds for every $\xi \in P$. $\square$

MAIN THEOREM

Theorem 2. *Let $(\mathcal{A}, \mathcal{B}, \mathcal{F}, d, \tau)$ be a canonical fusion bialgebra. Every minimizer of the Donoho–Stark uncertainty principle is a bishift of a biprojection. The same conclusion holds after any normalized gauge transformation (3).*

Proof. Multiplying the minimizer by a nonzero scalar does not affect the claim. As $\mathcal{A}$ is abelian and the source’s equality theorem makes w a multiple of a partial isometry, we may write

$$(4) \quad w = \sum_{j \in J} \varepsilon_j p_j, \quad |\varepsilon_j| = 1, \quad D := \mathcal{S}(w) = \sum_{j \in J} d_j^2.$$

Put

$$a = \mathcal{F}(w) = \sum_{j \in J} \varepsilon_j d_j x_j.$$

Extremality gives $\|a\|_\infty = \|w\|_1 = D$. Since a is a multiple of a partial isometry, there is a partial isometry V such that

$$(5) \quad a = DV.$$

Write $E = V^*V$ and $Q = VV^*$. Plancherel and traciality give

$$(6) \quad D = \|w\|_2^2 = \|a\|_2^2 = D^2\tau(E), \quad \tau(Q) = \tau(E) = D^{-1}.$$

For a unit vector $\xi \in E$, equality in

$$D = \|a\xi\| \leq \sum_{j \in J} d_j \|x_j \xi\| \leq \sum_{j \in J} d_j^2 = D$$

is termwise. Applying the same argument to a^* on Q yields

$$(7) \quad x_j E = d_j \bar{\varepsilon}_j V, \quad x_j^* Q = d_j \varepsilon_j V^* \quad (j \in J).$$

Step 1: the Fourier stabilizer is a fusion subalgebra. Let H be the set of indices s for which there is a phase α_s such that

$$(8) \quad x_s Q = d_s \alpha_s Q, \quad x_s^* Q = d_s \overline{\alpha_s} Q.$$

The identity belongs to H , and (8) makes H invariant under $s \mapsto s^*$, with $\alpha_{s^*} = \overline{\alpha_s}$.

If $s, t \in H$, then

$$x_s x_t Q = d_s d_t \alpha_s \alpha_t Q.$$

Expand $x_s x_t$ in the $\mathbb{R}_+$ -basis. Equation (1) says that the sum of the coefficient weights is $d_s d_t$, so Lemma 1 shows that every constituent r lies in H and has $\alpha_r = \alpha_s \alpha_t$. Applying the same argument to the adjoint product supplies the second equation in (8). Thus H is a fusion subalgebra basis.

For $j, k \in J$, equations (7) give

$$(9) \quad x_j x_k^* Q = d_j d_k \varepsilon_k \bar{\varepsilon}_j Q.$$

Lemma 1, first for this product and then for its adjoint, shows that every constituent of $x_j x_k^*$ belongs to H .

Step 2: the stabilizer and the support have the same dimension. Every product occurring in aa^* is supported in H . Conversely, for $s \in H$, orthonormality of the fusion basis, (8), and (6) give the x_s -coefficient of $aa^* = D^2 Q$:

$$\tau(x_s^* aa^*) = D^2 \tau(x_s^* Q) = D d_s \overline{\alpha_s}.$$

Consequently

$$(10) \quad aa^* = DR_\alpha, \quad R_\alpha := \sum_{s \in H} \overline{\alpha_s} d_s x_s, \quad R_\alpha = DQ.$$

Put $D_H = \sum_{s \in H} d_s^2$. Fusion closure, phase multiplicativity, and Frobenius reciprocity imply

$$R_\alpha^2 = D_H R_\alpha.$$

For completeness, the coefficient of x_r , $r \in H$, is

$$\overline{\alpha_r} \sum_{s, t \in H} d_s d_t N_{s, t}^r = \overline{\alpha_r} \sum_{s \in H} d_s \sum_{t \in H} d_t N_{s^*, r}^t = D_H \overline{\alpha_r} d_r.$$

On the other hand, $R_\alpha = DQ$ and $Q^2 = Q$ give $R_\alpha^2 = DR_\alpha$. Hence

$$(11) \quad D_H = D.$$

Step 3: the support is an H -coset. The relations (7) characterize J . Indeed, if an index r and a phase ε_r satisfy both equations in (7), then

$$\tau(x_r^*a) = D\tau(x_r^*QV) = Dd_r\varepsilon_r\tau(E) = d_r\varepsilon_r \neq 0,$$

so r occurs in the fusion-basis expansion of a and therefore $r \in J$.

Now let $s \in H$ and $j \in J$. Using (8) and (7),

$$x_sx_jE = d_sd_j\alpha_s\bar{\varepsilon}_jV, \quad x_j^*x_s^*Q = d_sd_j\bar{\alpha}_s\varepsilon_jV^*.$$

Apply Lemma 1 to the product and its adjoint. Every constituent r of x_sx_j satisfies (7) with $\varepsilon_r = \bar{\alpha}_s\varepsilon_j$, hence lies in J . Thus

$$(12) \quad HJ \subseteq J.$$

Step 4: the two shift equations. Define projections in $\mathcal{A}$ by

$$B_H = \sum_{s \in H} p_s, \quad P_J = \sum_{j \in J} p_j = \mathcal{R}(w).$$

The regular element of the fusion subalgebra H satisfies

$$\mathcal{F}(B_H) = \sum_{s \in H} d_s x_s = DR_H, \quad R_H := D^{-1} \sum_{s \in H} d_s x_s,$$

where R_H is a projection. Thus B_H is a biprojection and $d(B_H) = D = d(P_J)$.

Equation (12) shows that $\mathcal{F}(B_H)\mathcal{F}(P_J)$ has no constituents outside J . For $r \in J$, its x_r -coefficient is, using (2) and again (12),

$$\sum_{s \in H, j \in J} d_sd_jN_{s,j}^r = \sum_{s \in H} d_s \sum_{j \in J} d_j N_{s^*,r}^j = \sum_{s \in H} d_s^2 d_r = Dd_r.$$

Therefore

$$(13) \quad B_H * P_J = DP_J = d(B_H)P_J,$$

so P_J is a right shift of B_H .

Equations (10) and (11) say

$$(14) \quad Q = D^{-1} \sum_{s \in H} \bar{\alpha}_s d_s x_s, \quad \tau(Q) = \tau(R_H) = D^{-1}.$$

Use the source's dual convolution formula

$$X *_B Y = (\mathcal{F}(\mathcal{F}^{-1}(Y^*) \diamond \mathcal{F}^{-1}(X^*)))^*.$$

Since $\mathcal{F}^{-1}(R_H) = D^{-1} \sum_{s \in H} p_s$ and $\mathcal{F}^{-1}(Q) = D^{-1} \sum_{s \in H} \bar{\alpha}_s p_s$, coordinatewise multiplication in $\mathcal{A}$ gives

$$(15) \quad Q *_B R_H = D^{-1}Q = \tau(R_H)Q.$$

Thus Q is a shift of $\mathcal{R}(\mathcal{F}(B_H)) = R_H$.

Finally, P_J supports w and Q is the range projection of $\mathcal{F}(w)$. Hence

$$w \diamond P_J = w, \quad Q\mathcal{F}(w) = \mathcal{F}(w),$$

and therefore

$$(16) \quad w = \mathcal{F}^{-1}(Q) * (w \diamond P_J).$$

Together with (13) and (15), this is exactly the definition of a bishift of the biprojection B_H .

The gauge statement was verified above (3). $\square$

WHAT REMOVES THE DUAL YOUNG HYPOTHESIS

The source’s conditional theorem follows the positive-projection route and uses dual Young’s property to control convolution on $\mathcal{B}$. The proof above does not try to make a dual convolution positive. It instead extracts the fusion subalgebra directly from the norm-saturating action of the original $\mathbb{R}_+$ -basis on the Fourier range projection. Positivity is used only for the structure constants $N_{j,k}^r$, where it is part of the definition of a fusion bialgebra and is available without any dual assumption.

VERIFICATION, LIMITATIONS, AND NOVELTY CHECK

The argument is algebraic and applies to arbitrary finite rank and to noncommutative $\mathcal{B}$. It does not assume simplicity, integrality, subfactorizability, self-duality, the dual Schur property, or dual Young’s property. The only normalization issue is explicit: for an unrestricted two-parameter gauge the sharp support-product constant is rescaled, so the literal constant-one formulation should first be put in canonical (or product-one) gauge.

The proof was audited against the source definitions. The retained rank-three symbolic checker and a finite-group checker for noncommutative group algebras both pass; these are sanity checks, not proof ingredients.

For novelty, on 22 July 2026 we searched the run indexes and arXiv using the source identifier, the exact wording of Question 6.15, and the core terms above together with *dual Young property*. No later answer was found. This bounded check is not a claim of publication priority.

HUMAN-REVIEW RECOMMENDATION

The most valuable review is concentrated in four places: the termwise saturation in Lemma 1; the need for both stabilizer equations in (8); the coefficient computation proving $D_H = D$; and the order of the two inverse Fourier factors in (15). If these pass, the packet gives a full affirmative solution in the source’s normalized setting.

REFERENCES

- [1] Z. Liu, S. Palcoux, and J. Wu, *Fusion bialgebras and Fourier analysis*, Adv. Math. **390** (2021), 107905; arXiv:1910.12059.